CIRC HQ STUDENT HANDOUT | GRADE 5

Maze Coders

Goal: Write an efficient algorithm that moves a token through a grid maze without entering blocked cells.

35 minutes | 8 steps | Computer Science and Programming | Playbook B meeting 15, grade 5 path

Ask | Plan | Build | Test | Improve

You are done when

Submit the final command count and circle the command that fixed the first bug.

You will use

- Grid maze cards
- Tokens
- Command cards
- Pencils
- Transparent sleeves

Steps

35 minutes total. Your teacher says when to move on.

1 Step 1: Get ready — 3 min

- Place one maze card in the sleeve.
- Put the token on start.

2 Step 2: Token safety — 3 min

- Keep the token on the grid.
- Pick up any dropped token at once.

3 Step 3: Map the route — 5 min

- Mark the goal.
- Mark each blocked cell.
- Trace one possible route.

4 Step 4: Write the code — 7 min

- Write one command per box.
- Finish the full route before testing.

5 Step 5: Partner test — 6 min

- Trade code with a partner.
- Run each command exactly.
- Mark the first error.

6 Step 6: Debug the route — 5 min

- Change the first wrong command.
- Remove one extra command if possible.
- Run the route again.

7 Step 7: Reset the maze — 3 min

- Return the token.
- Erase the sleeve.

8 Step 8: Exit check — 3 min

- Write the number of final commands.
- Circle the command you changed.

What to turn in

Your final command count and the command that fixed the first bug.

No computer? Another way to do it

This lesson is already device free.

Finished early? Next challenge

Shortest Safe Route: Find a second valid route and prove which one uses fewer commands.

Stay safe

Keep all work on the grid surface and keep small tokens away from mouths and floor traffic.

Stuck? Try this order

1. Read the step again, out loud if you can.
2. Ask a partner at your table.
3. Raise your hand. Keep working on the next part while you wait.